ALEXANDRIA QUANTUM HACKATHON 2026 • TEAM PARACHUTE

Quantum Hardware Design & Topology Optimization

Simulation, Routing, DRC Verification, and Comparative Analysis of Five-Qubit Layouts

Abstract. *We design, optimize, and compare two five-qubit superconducting transmon layouts—a linear nearest-neighbor chain and a hub-centered star—implemented in Qiskit Metal with RoutePathfinder couplers. Both topologies are placed on an identical* 12×10 *mm footprint under one shared design-rule set, so topology is the only independent variable. Graph-level connectivity analysis is combined with physical layout optimization, in which qubit pitch (chain) or hub radius (star) is swept and every candidate is filtered through a hard Design-Rule-Check (DRC) gate before ranking by total coupler routing length. After correcting the chain's connection-pad wiring, the optimized chain (1.2 mm pitch) achieves a total routing length of 1.400 mm versus 3.356 mm for the optimized star (1.5 mm radius): a* $2.4\times$ *shorter routing budget, a* $2.4\times$ *shorter worst-case coupler, and a* $2.2\times$ *larger route-to-route clearance, with both designs crossing-free and fully DRC-compliant. We recommend fabricating the linear chain, with a documented workload-dependent exception favoring the star for hub-centered interaction patterns.*

Project design snapshot

Qubits	5 × TransmonPocket	**Opt. variable**	Chain pitch / star hub radius
Topologies	Linear chain and star	**Routing**	Qiskit Metal RoutePathfinder
Chip footprint	12×10 mm (shared)	**Baseline**	2.5 mm pitch / radius
CPW geometry	10 µm trace / 6 µm gap	**Final designs**	Chain 1.2 mm / Star 1.5 mm

1 Introduction and Objectives

This project develops and compares two five-qubit superconducting layouts: a linear nearest-neighbor *chain* and a hub-centered *star*. Both are implemented in Qiskit Metal using `TransmonPocket` qubits and `RoutePathfinder` couplers. The study deliberately couples graph-level topology analysis with physical layout optimization, routing verification, manufacturability rules, and EM-aware considerations.

The goal is not to claim that one topology is universally superior. Instead, the comparison identifies the layout that offers the strongest *physical* implementation for the tested design constraints, while keeping the workload-dependence of topology selection explicit.

Main objectives

- Build both topologies on identical physical real estate.
- Define reproducible qubit, CPW, spacing, and edge keep-out rules.
- Optimize physical spacing using a *hard* DRC filter.
- Route all couplers and verify crossings and route clearances.
- Compare graph, routing, and manufacturability metrics.
- Select a final topology and state the workload-dependent exception.

2 Baseline Design and System Model

2.1 Chip footprint and spacing

Both candidates use a standardized 12×10 mm footprint, which removes footprint selection as a confounding factor in the comparison. The baseline pitch/radius is 2.5 mm. At this spacing the outer chain qubits are centered at ± 5.0 mm; the pocket-plus-pad half-width is 0.425 mm, leaving ≈ 0.575 mm to the chip edge—above the 0.5 mm keep-out requirement. The broader exploratory sweep is intentionally allowed to include *infeasible* values; these are rejected by the hard DRC filter before optimization, so an oversized search range can never be selected as a final design.

2.2 Qubit and CPW geometry

Table 1 lists the fixed geometry and design rules used for both topologies. These are the assumptions encoded directly in the layout scripts.

2.3 Coupling model

A coupling map is treated as a graph in which qubits are nodes and physical couplers are edges. Both layouts contain five qubits and four couplers, but their connectivity differs. The topology therefore changes the number of hops required for interactions and can introduce different routing and SWAP overheads.

Table 1. Fixed qubit / CPW geometry and design rules.

Parameter	Value
Pad width	425 µm
Pad height	90 µm
Pocket width	650 µm
Pocket height	650 µm
Nominal CPW trace	10 µm
Nominal CPW gap	6 µm
Absolute CPW floor	8 µm / 4 µm
Chip-edge keep-out	$\geq$ 0.5 mm
Minimum qubit edge gap	$\geq$ 0.3 mm
Route fillet radius	50 µm
Straight lead-in/out	100 µm

2.4 Implementation note: connection pads

Connection pads are constrained to pocket corners. For the chain, each qubit is connected using the *near* pad facing its neighbor. This correction is important: the earlier far-pad wiring forced unnecessary detours and produced misleadingly long routes (see Section 5).

3 Topology Exploration

The two candidate coupling graphs are shown in Fig. 1. Their graph metrics—computed with NetworkX—directly motivate the topology trade-off.

3.1 Linear chain

The chain connects Q1–Q2–Q3–Q4–Q5, giving maximum degree 2, graph diameter 4, and average path length 2.00. Its sparse, uniform connectivity is a natural match for nearest-neighbor workloads such as chain-structured QAOA or Trotterized Hamiltonians.

3.2 Star

The star uses one central hub connected to four outer qubits, giving maximum degree 4, diameter 2, and average path length 1.60. This is attractive for hub-centered patterns such as GHZ-state preparation. Importantly, the star should *not* be described as automatically lowering circuit depth for GHZ preparation: two-qubit gates that share the central qubit must be serialized. Its advantage is the direct *topological* match, not a universal speedup. The hub can also become a physical crowding and crosstalk concern.

3.3 Engineering trade-off

- **Chain:** simpler connectivity, lower hub crowding, and better physical routing after correcting the pin wiring; but distant qubits are not directly connected, so nonlocal interactions may require more SWAPs.
- **Star:** shorter graph distances and direct hub-to-spoke interactions; but four couplers converge on one hub, increasing layout crowding and crosstalk exposure.

Table 2. Graph-level comparison of the two topologies.

Property	Chain	Star
Qubits / couplers	5 / 4	5 / 4
Maximum degree	2	4
Diameter	4	2
Average path length	2.00	1.60
Natural workload	Local / 1D	Hub-centered
Crossings (final)	0	0

Table 3. Routing-length optimization results (objective: total coupler length).

Topo.	Baseline	Grid optimum	Red.
Chain	6.600 mm	1.400 mm @ 1.2 mm pitch	78.8%
Star	9.012 mm	3.356 mm @ 1.5 mm radius	62.8%

At the 2.5 mm baseline, the physical cost of the denser star connectivity is already visible: total routing length is 6.600 mm (chain) versus 9.012 mm (star), and the longest route is 1.650 mm versus 2.253 mm.

4 Physical Optimization and EM-Aware Routing

4.1 Optimization method

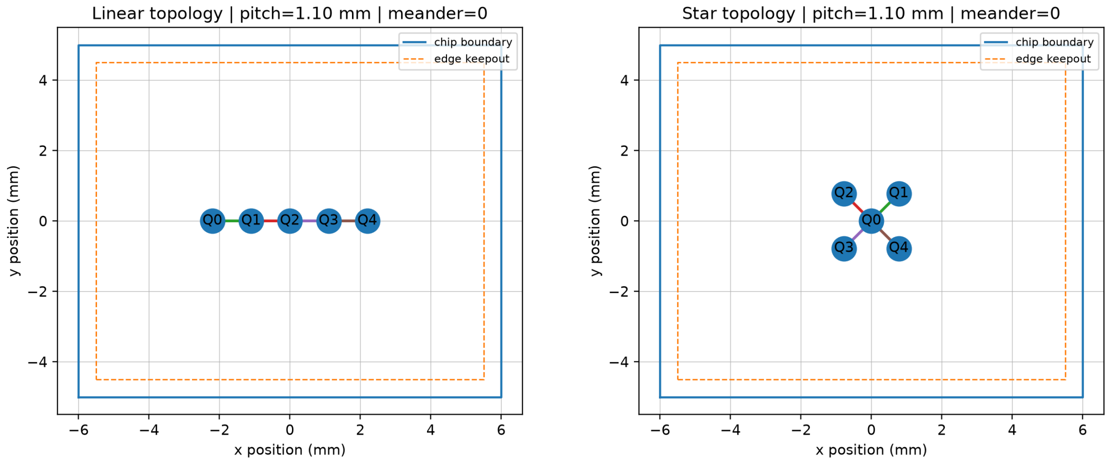
The optimization objective is *total coupler routing length*. The variable is qubit pitch (chain) or hub radius (star). Candidate layouts are first checked against hard DRC constraints; only feasible designs are ranked. This prevents an invalid but short layout from being selected as the optimum. Figure 2 shows the built route length together with the number of geometry-rule violations across the sweep—directly exposing the feasible design window.

4.2 Optimization results

Table 3 reports the grid optima. A finer continuous boundary search (Fig. 3) indicates a slightly tighter feasible point, $\approx$ 1.15 mm pitch (chain) and $\approx$ 1.33–1.35 mm radius (star). The grid optima are retained as the reported final designs because they are explicit, DRC-passing search results.

4.3 Final geometry, routing, and crossings

All couplers are routed with `RoutePathfinder` using 50 µm bend fillets and 100 µm straight leads. The final chain (1.2 mm pitch) and final star (1.5 mm radius) are both crossing-free, and neither requires an airbridge (Fig. 4).



(a) Linear chain: Q0–Q1–Q2–Q3–Q4 (b) Star: central hub Q0 with four spokes

Figure 1. Candidate coupling graphs on the shared 12 × 10 mm footprint with the 0.5 mm edge keep-out (dashed). The chain has maximum degree 2 (diameter 4, average path length 2.00); the star has maximum degree 4 (diameter 2, average path length 1.60). These graph metrics are verified in code and summarized in Table 2.

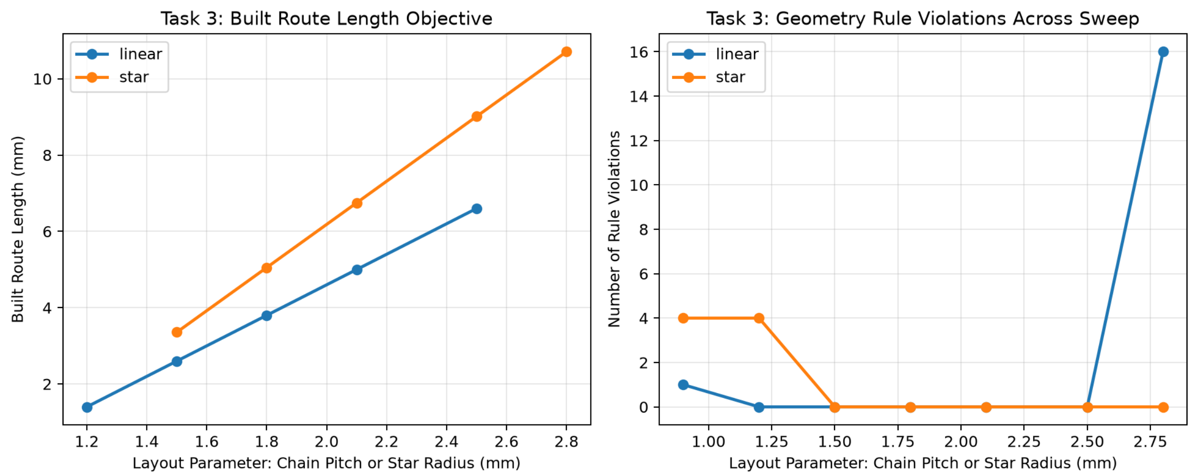


Figure 2. Parameter sweep over the layout variable (chain pitch / star radius). **Left:** built route length grows monotonically with spacing for both topologies, and the chain is uniformly shorter. **Right:** geometry-rule violations reveal the feasible band—very tight spacings and the widest spacings both violate DRC (e.g. the linear point at ∼2.8 mm spikes to 16 violations, and tight sub-1.2 mm layouts fail), whereas the mid-range is clean. This plot verifies the code assumption that routing success alone is insufficient: DRC is a hard gate.

Table 4. Final-design route metrics.

Final design	Longest coupler	Min. clearance
Chain — 1.2 mm pitch	0.350 mm	0.850 mm
Star — 1.5 mm radius	0.839 mm	0.390 mm

4.4 Route clearance

Table 4 quantifies the routing outcome. The chain combines shorter routes with substantially larger separation between routes; the star's hub geometry produces tighter routing and greater crosstalk-proxy exposure.

4.5 EM interpretation

The final couplers are very short direct links rather than frequency-tuned distributed resonators. As a reference only, a 6 GHz quarter-wave line with $\varepsilon_{\text{eff}} = 6$ would be ≈ 5.10 mm long—much longer than either final coupler. Full electromagnetic extraction would be required for quantitative coupling and frequency characterization.

5 Design Rules, Validation, and Iteration

5.1 Final DRC verification

The project treats DRC as a hard feasibility gate rather than a soft score. Table 5 shows that both final designs pass every rule. Appendix B illustrates

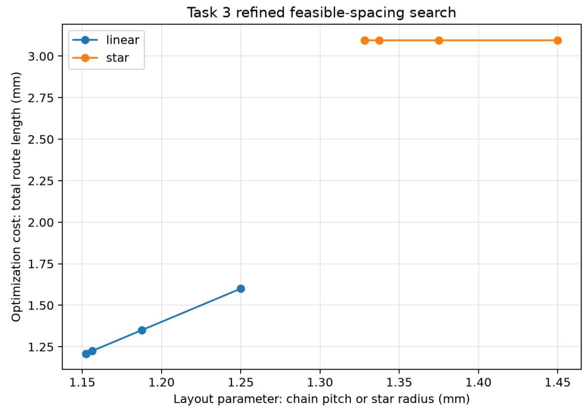


Figure 3. Refined feasible-spacing search near the DRC boundary. The chain reaches its shortest feasible total length at $\approx$ 1.15 mm pitch; the star's feasible points cluster near 1.33–1.45 mm radius at a much higher cost, confirming the chain's routing advantage persists at the continuous optimum.

Table 5. Final DRC verification for both selected designs.

Rule	Chain 1.2	Star 1.5
Min. edge gap ≥ 0.3 mm	0.350 **PASS**	0.462 **PASS**
CPW trace / gap rule	**PASS**	**PASS**
Chip-edge keep-out ≥ 0.5 mm	**PASS**	**PASS**
Crossings	0	0
Airbridges required	0	0

the pass/fail methodology.

5.2 Important design iterations

- Star connection pads were initially placed at compass directions that are invalid for `TransmonPocket`; diagonal outer-qubit placement solved the pin-geometry constraint.
- The routing-length metric was initially misinterpreted; the router's actual computed length was used instead of the meander target.
- Very tight chain spacing caused `RoutePathfinder` failure, establishing a practical lower search bound.
- A 1.0 mm chain pitch routed successfully but failed DRC—demonstrating why routing success alone is insufficient.
- Star routing became numerically unreliable below $\approx$ 0.9 mm radius; near-boundary results were cross-checked against DRC.
- Fillet warnings were confirmed cosmetic: no effect on DRC, routing lengths, or final geometry.
- The full Qiskit install conflicted with the Quantum Metal environment, so NetworkX was used for graph metrics.
- Both topologies were standardized to the same 12×10 mm footprint to make the final comparison fair.

5.3 Why the chain result changed

The chain was found to be using the *wrong* connection pads. Switching to near-pad connections reduced routing length from 18.657 mm to 6.600 mm at 2.5 mm pitch—a 65% reduction. This was not cosmetic: the original wiring forced `RoutePathfinder` to travel *around* qubit pockets instead of taking the direct neighboring connection. After correction, the chain became both shorter and better separated than the star, which is essential to interpreting the final recommendation.

6 Final Comparison and Recommendation

Table 6 consolidates the graph, routing, and manufacturability metrics for the two final designs.

Final recommendation

Fabricate the linear chain. With the corrected pin wiring it wins on all measured physical routing metrics—total routing length, longest/weakest-link length, and minimum route-to-route clearance—while avoiding the degree-4 hub crowding of the star.

- $\approx 2.4\times$ shorter total routing length.
- $\approx 2.4\times$ shorter longest coupler (weakest link).
- $\approx 2.2\times$ larger minimum route clearance.
- Maximum degree 2 avoids concentrating four couplers on one transmon.
- Both layouts are crossing-free and DRC-passing, so those do not distinguish the designs.

6.1 Workload-dependent exception

If the target workload is specifically hub-centered—for example, a design dominated by central-to-all-qubit interactions—the star remains the more natural architecture, because its coupling graph directly matches that interaction pattern. This may outweigh its physical routing disadvantages. The recommendation is therefore workload-conditional, not a claim that the chain is universally superior.

6.2 Limitations

- Based on simulated Qiskit Metal layouts and DRC checks, not fabricated hardware measurements.
- Full EM extraction is outside the scope of this comparison.
- Finite-size effects, fabrication variation, device-specific losses, and detailed crosstalk are not experimentally characterized.
- Continuous-optimum values are presented as refinement points; the selected designs use explicit grid-search results.

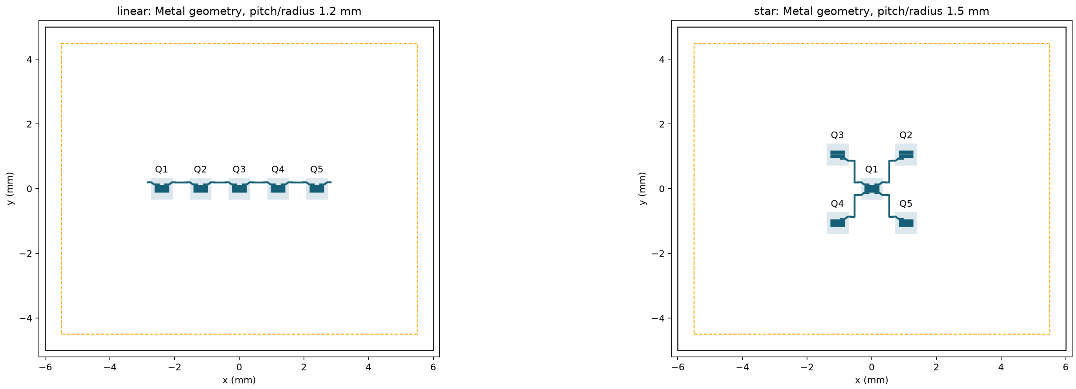


Figure 4. Final Qiskit Metal geometry: the linear chain at 1.2 mm pitch (left) and the star at 1.5 mm radius (right), both inside the 0.5 mm edge keep-out (dashed). Both are crossing-free and airbridge-free, verifying the routing assumptions used in the comparison.

Table 6. Head-to-head comparison of the two final, DRC-passing designs.

Property	Linear chain	Star	Comment
Footprint	12×10 mm	12×10 mm	Shared, identical
Maximum degree	2	4	Star concentrates 4 couplers on hub
Diameter / avg. path	4 / 2.00	2 / 1.60	Star has shorter graph distances
Final spacing	1.2 mm pitch	1.5 mm radius	Grid optima (DRC-passing)
Total routing length	**1.400 mm**	3.356 mm	Chain $\approx 2.4\times$ shorter
Longest coupler	**0.350 mm**	0.839 mm	Chain $\approx 2.4\times$ shorter
Min. route clearance	**0.850 mm**	0.390 mm	Chain $\approx 2.2\times$ larger
Crossings / airbridges	0 / 0	0 / 0	Non-differentiating
DRC	**PASS**	**PASS**	Both compliant

A Broader Exploratory Sweep

Beyond the two carried-forward candidates, a broader normalized screening also included a ring reference (Fig. 5). The normalized 1-D proxy does not penalize the ring's additional wrap-around coupler; detailed physical optimization and DRC (Fig. 2) were therefore carried out for the two four-coupler candidates, chain and star. A two-dimensional sweep over pitch and meander density (Fig. 6) confirms that cost is minimized at tight, low-meander configurations for every topology.

B DRC Validation Methodology

Figure 7 summarizes how the Design-Rule Check is applied: geometry is built, then spacing, CPW width/gap, and chip-edge keep-out are checked against one shared rule set. A normal baseline passes, while an intentionally tight layout fails—demonstrating that the checker actively catches violations.

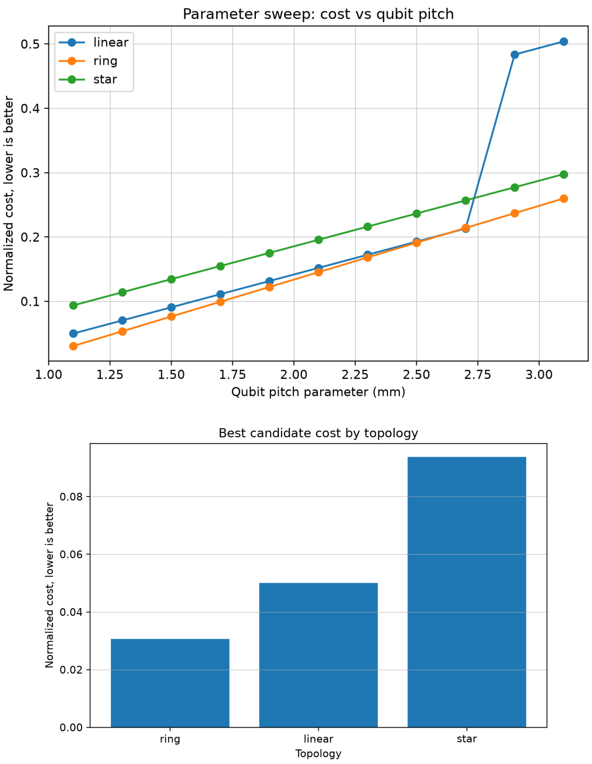


Figure 5. Exploratory screening. **Top:** normalized cost versus qubit pitch for the linear, ring, and star candidates. **Bottom:** best-candidate normalized cost by topology (lower is better).

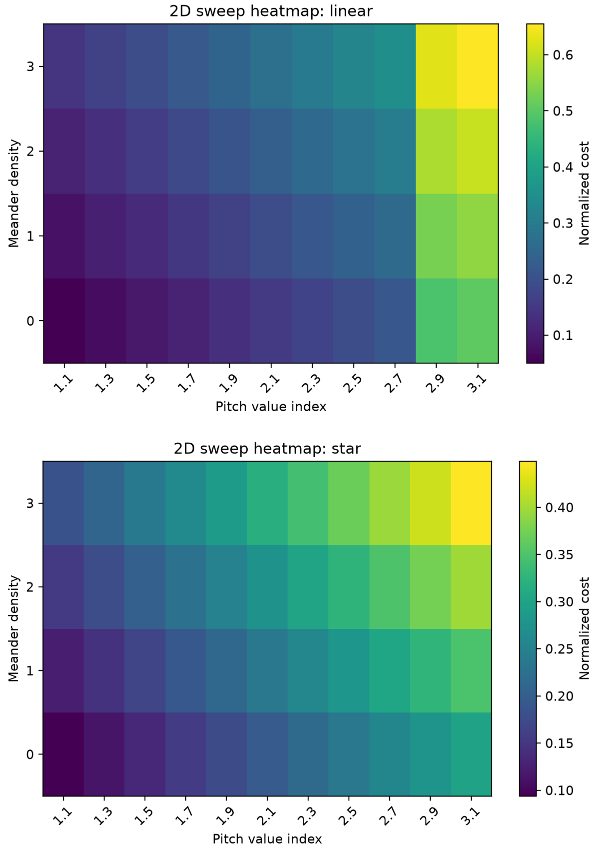


Figure 6. Two-dimensional sweep heatmaps (pitch × meander density) for the linear (top) and star (bottom) topologies. Cost is lowest at tight pitch and low meander density, and grows toward wide/high-meander corners.

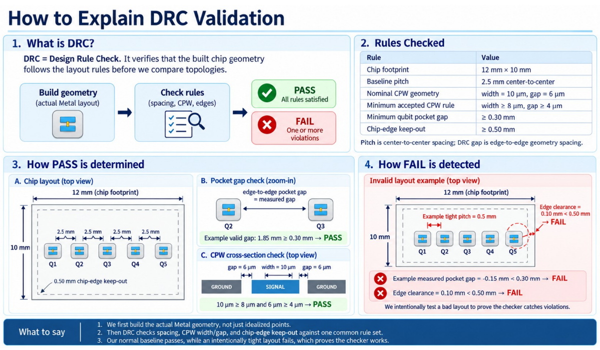


Figure 7. DRC validation flow and worked pass/fail examples used to validate the hard-gate assumption in the optimization loop.